

NUMBER PATTERNS

Short Revision Notes — Analyze Level (Break Information into Parts and Identify Relationships)

1. Breaking Down the General Term

Component	What It Represents	Example: $T_n = 4n + 7$
T_n	The value of the n-th term	$T_5 = 27$
n	The position of the term	For 5th term, $n = 5$
Coefficient of n (4)	The common difference (d)	$d = 4$
Constant term (7)	The value when $n = 0$ (starting point)	When $n=0$, $T = 7$
First term (T_1)	$T_1 = d \times 1 + c$	$T_1 = 4(1) + 7 = 11$

The Relationship: $T_n = dn + c$ where: d = common difference, $c = T_1 - d$

2. Relating the Sequence to Its General Term

Sequence	d	c	T_n	Check T_1	Check T_2
5, 8, 11, 14...	3	2	$3n + 2$	$3(1)+2=5$ ✓	$3(2)+2=8$ ✓
1, 7, 13, 19...	6	-5	$6n - 5$	$6(1)-5=1$ ✓	$6(2)-5=7$ ✓
22, 19, 16, 13...	-3	25	$25 - 3n$	$25-3=22$ ✓	$25-6=19$ ✓

Relationship Pattern: As n increases by 1, T_n changes by d . If $d > 0 \rightarrow$ sequence INCREASES. If $d < 0 \rightarrow$ sequence DECREASES. If $d = 0 \rightarrow$ sequence CONSTANT.

3. Comparing Different Sequences

Feature	Type 1 (Arithmetic)	Type 2 (Non-Arithmetic)
Rule	Add/subtract same number each time	Multiply, divide, or varied rule
Example	2, 4, 6, 8, 10...	2, 4, 8, 16, 32...
Common Difference	YES ($d = 2$)	NO
General Term	$T_n = dn + c$	$T_n = 2^n$ or other rule

Comparison Analysis:

Sequence	Common Difference?	Pattern Type
3, 3, 3, 3, 3...	Yes ($d = 0$)	Constant
5, 8, 11, 14...	Yes ($d = 3$)	Increasing
20, 15, 10, 5...	Yes ($d = -5$)	Decreasing
1, 2, 4, 8, 16...	No	Multiplying by 2
1, 1, 2, 2, 3, 3...	No	Alternating groups

4. Analyzing Term Position Problems

Problem	Method	Analysis Steps
Is 75 a term of $T_n = 4n - 3$?	Set $4n - 3 = 75$	$4n = 78 \rightarrow n = 19.5$
Check: Is n a whole number?		19.5 is NOT a whole number

Problem	Method	Analysis Steps
Conclusion: 75 is NOT a term		✓
Is 0 a term of $T_n = 56 - 4n$?	Set $56 - 4n = 0$	$n = 14$
Check: Is n a whole number?		14 IS a whole number
Conclusion: 0 IS a term (14th term)		✓

5. Breaking Down “Which Term” Problems

Question	Equation	Solve for n	Result
Which term is 123?	$2n + 3 = 123$	$n = 60$	60th term
Which term is -15?	$50 - 7n = -15$	$n = 65/7 \approx 9.29$	NOT a term
Which term is 100?	$5n - 2 = 100$	$n = 20.4$	NOT a term
Which term is 37?	$3n + 1 = 37$	$n = 12$	12th term

Analysis Pattern: If n is a POSITIVE INTEGER $\rightarrow$ the value IS a term. If n is NOT a positive integer $\rightarrow$ the value IS NOT a term.

6. Identifying Relationships in Matchstick Patterns

Figure	Matchsticks	Increase	Analysis
1	3	-	$T_1 = 3$
2	5	+2	$T_2 = 5$
3	7	+2	$T_3 = 7$
4	9	+2	$T_4 = 9$

Relationship Found: Common difference = 2, First term = 3, General term: $T_n = 2n + 1$

Check relationship: $T_1 = 2(1) + 1 = 3$ ✓ $T_2 = 2(2) + 1 = 5$ ✓ $T_3 = 2(3) + 1 = 7$ ✓

7. Comparing Positive and Negative Common Differences

Feature	d = Positive	d = Negative
Direction	Sequence goes UP	Sequence goes DOWN
Example	3, 6, 9, 12...	12, 9, 6, 3...
T_n Form	$T_n = 3n + c$	$T_n = -3n + c$
$T_1 = 5$	$T_n = 5 + (n-1)3$	$T_n = 5 + (n-1)(-3)$
T_{20}	$5 + 19(3) = 62$	$5 - 19(3) = -52$
Terms get...	Larger and larger	Smaller and smaller

8. Analyzing the Relationship Between Terms

Relationship	Formula	Example ($T_n = 2n + 3$)
Term to next term	$T_{n+1} = T_n + d$	$T_7 = 17, T_8 = 19 (+2)$
Term to previous term	$T_{n-1} = T_n - d$	$T_7 = 17, T_6 = 15 (-2)$
Sum of two terms	$T_n + T_m$	$T_2 + T_3 = 7 + 9 = 16$
Difference of two terms	$T_n - T_m = (n-m)d$	$T_5 - T_2 = 13 - 7 = 6$

9. Relating General Term to Different n Values

n	T_n	Analysis
0	$T_0 = c$	This is the y-intercept (not a term)
1	$T_1 = d + c$	First term
2	$T_2 = 2d + c$	Second term
3	$T_3 = 3d + c$	Third term
n	$T_n = nd + c$	General term

Pattern Analysis: The value INCREASES by d each time n increases by 1. The value at n=0 is c (the constant). The first term is d more than the constant.

10. Breaking Down “Common Difference” From Terms

Given Terms	Analysis to Find d
$T_1 = 5, T_2 = 11$	$d = 11 - 5 = 6$
$T_1 = 10, T_4 = 22$	$3d = 22 - 10 = 12 \rightarrow d = 4$
$T_2 = 15, T_5 = 33$	$3d = 33 - 15 = 18 \rightarrow d = 6$
$T_3 = 20, T_7 = 40$	$4d = 40 - 20 = 20 \rightarrow d = 5$

General Relationship: $T_m - T_n = (m-n)d$

11. Analyzing Negative Terms in a Sequence

Question	Method	Analysis
When do terms become negative?	Find when $T_n < 0$	$T_n = 50 - 7n$
	$50 - 7n < 0$	$50 < 7n$
	$n > 7.14$	$n \geq 8$
Check:	$T_7 = 50 - 49 = 1$ (positive)	$T_8 = 50 - 56 = -6$ (negative)
Conclusion:	Terms after the 7th term are negative ✓	

Pattern Analysis: Terms decrease by 7 each step. Starting at 50, it takes 8 steps to become negative. Terms near the boundary: 50, 43, 36, 29, 22, 15, 8, 1, -6, -13...

12. Analyzing Multiple Sequences (Mixed Patterns)

Sequence Pair	Analysis	Relationship
5, 8, 11, 14...	$d = 3$	-
5, 11, 17, 23...	$d = 6$	Second sequence grows TWICE as fast
5, 5, 5, 5...	$d = 0$	Constant
5, 2, -1, -4...	$d = -3$	Opposite direction to first

Comparison Analysis: Same first term → different common differences. Larger d → faster increase. Negative d → decreasing sequence.

13. Analyzing Sequences by Term Position

Position (n)	Term Value	Increase (d)	Observation
1	3	-	-
2	7	+4	$d = 4$
3	11	+4	$d = 4$
4	15	+4	$d = 4$
5	19	+4	$d = 4$

Analysis: Constant increase of 4 confirms it's a common difference sequence. $T_n = 4n - 1$. Check: $n=1 \rightarrow 4(1)-1 = 3$ ✓

14. Checking Relationships in Word Problems

Problem	Given	Find	Relationship
Runner: Day 1 = 500m, Day 2 = 600m	$T_1=500$, $d=100$	T_{20}	$T_{20} = 500 + 19(100) = 2400\text{m}$
Seats: Row 1 = 9, Row 2 = 12	$T_1=9$, $d=3$	T_{15}	$T_{15} = 9 + 14(3) = 51$ seats
Salary: Year 1 = 25000, Year 2 = 27400	$T_1=25000$, $d=2400$	T_5	$T_5 = 25000 + 4(2400) = 34600$

15. Self-Test — Analysis Questions

No.	Question	Answer
1	What does the coefficient of n represent in T_n ?	Common difference
2	If $T_n = 5n - 3$, what is T_1 ?	2
3	How can you tell if a sequence is decreasing from T_n ?	The coefficient of n is negative
4	If $T_2 = 13$ and $T_5 = 25$, what is d ?	4
5	Is 76 a term of $T_n = 4n + 8$?	$4n+8=76 \rightarrow n=17 \rightarrow \text{YES}$
6	What does the constant term in T_n represent?	The value when $n=0$
7	If $T_n = 3n + 2$, what is the relationship between T_3 and T_4 ?	$T_4 = T_3 + 3$
8	What type of sequence is 7, 11, 15, 19?	Increasing with $d=4$
9	What type of sequence is 8, 5, 2, -1?	Decreasing with $d=-3$
10	When do terms become negative in $T_n = 40 - 5n$?	$n > 8$ (after 8th term)

16. Analysis Summary

SEQUENCE ANALYSIS CHECKLIST

- ✓ Find the common difference (d)
- ✓ Identify first term (T_1)
- ✓ Determine if d is positive, negative, or zero
- ✓ Write the general term: $T_n = T_1 + (n-1)d$
- ✓ Check if a value is a term: solve for n (must be whole number)
- ✓ Predict term behavior: increasing, decreasing, or constant
- ✓ Find term position: $n = (T_n - T_1) \div d + 1$
- ✓ Compare sequences: same d ? different d ? same T_1 ?
- ✓ Analyze real-world patterns: distance, seating, money, shapes

Revision Tip: When analyzing, always ask: "What does each part of T_n tell me?" Breaking it down makes everything clearer!